

# applied acoustics

underwater technology



## **E4947** **LF Transponder / Altimeter** **Board Set**



Registration No. QU180081a  
ISO9001:2015

## Revision History

| Issue | Change No. | Reason for change                                      | Date     |
|-------|------------|--------------------------------------------------------|----------|
| A     | -          | E4947 commands added. Pinger channels 0B & 10 removed. | 24/02/20 |
| B     | -          | Updates to Appendix section                            | 03/03/20 |
| 1     | -          | First Issue                                            | 01/04/20 |

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These written instructions must be followed fully for reliable and safe operation of the equipment that this manual refers to. Applied Acoustic Engineering Ltd cannot be held responsible for any issues arising from the improper use or maintenance of equipment referred to in this manual or failure of the operator to adhere to the instructions laid out in this manual. The user must be familiar with the contents of this manual before use or operation.



## 1. Introduction

Thank you for choosing Applied Acoustic Engineering as one of your subsea equipment suppliers. We hope you will enjoy many years of reliable operational use from our products.

The AAE Model SPC-5414-5000 LF Transponder / Altimeter board set consists of a combined transponder / pinger / altimeter, suiting a variety of underwater positioning and relocation tasks.

There is facility for operation with the following systems:-

|                        |                                 |
|------------------------|---------------------------------|
| Simrad HPR 1507 series | 5 Channels.                     |
| Simrad HPR4 LF series  | 30 Channels.                    |
| ORE Trackpoint LF      | 34 Channels.                    |
| Other Systems .....    | Which are frequency compatible. |

The transponder / altimeter board set has the capability to operate on 144 different transmit and receive frequency combinations referred to as 'channels', selected by two rotary hex switches mounted on the board.

The transponder / altimeter board set operates as follows:-

- a) Acoustically triggered transponder (requires one or two acoustic signals before replying).
- b) Free running pinger.
- c) Altimeter.

The transponder / altimeter board set is specifically for use with the AAE T154 transducer.

## 2. Getting Started

On delivery of your transponder board set, the following should be included:

- SPC-5414-5000 Transponder / Altimeter Board (complete with MkIII 'processor board).
- AAE T154 Transducer.
- SPC-5414-3000 external inductor (with 300mm wires fitted for convenience).
- SPC-5414-4001 external capacitor (with 300mm wires fitted for convenience).
- SPC-5414-3001 external transformer (with 300mm wires fitted for convenience).
- Transponder Switch Control Cable assembly.
- LF Transponder / Altimeter Board manual.

As indicated above, specific components to be mounted external to the transponder board have wiring factory-fitted, to be cut to the required lengths after customer mounting.

The inductor is supplied with a neoprene disc for mounting.



Note: supplied external components are specifically for use with model SPC-5414-5000 transponder / altimeter board and AAE T154 transducer.

### **3. System Assembly**

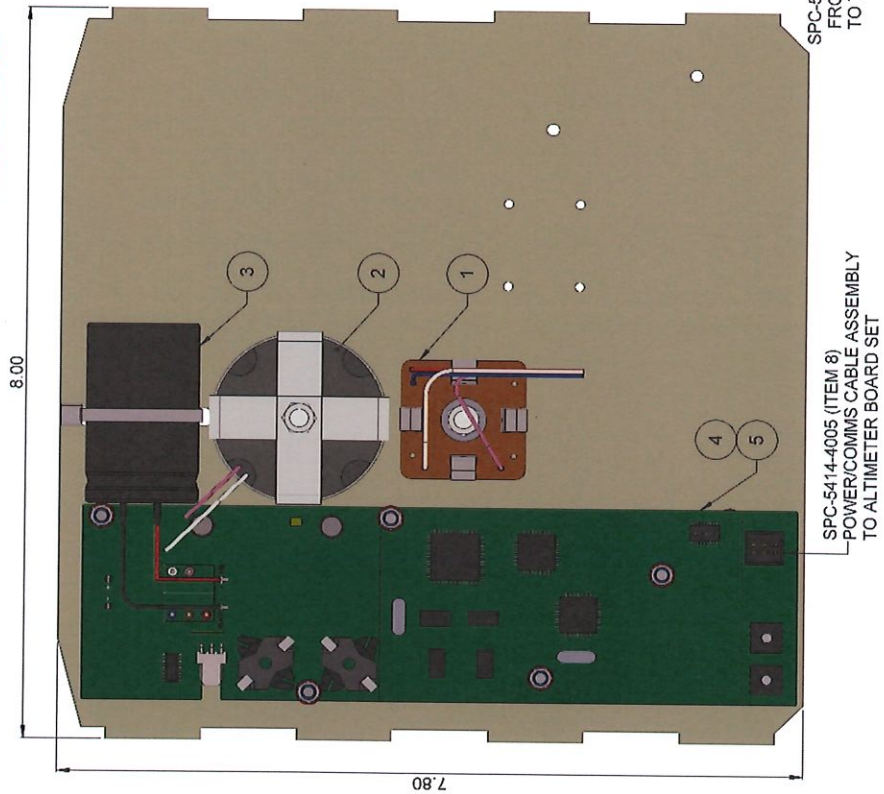
#### **Board Layout and Mounting**

AAE Ref: SPC-5414-7026-1

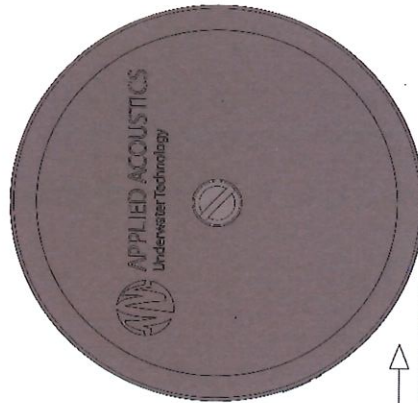
For detailed board and fixing point dimensions, refer section 9 'Appendix'.

| ITEM | QTY. | DESCRIPTION                                                          | VALUE                   | AAE STOCK CODE |
|------|------|----------------------------------------------------------------------|-------------------------|----------------|
| 1    | 1    | ALTIMETER EXTERNAL TRANSFORMER (15V)                                 |                         | SPC-5414-3001  |
| 2    | 1    | ALTIMETER ITC3013 EXTERNAL INDUCTOR                                  |                         | SPC-5414-3000  |
| 3    | 1    | EXTERNAL CAPACITOR                                                   |                         | SPC-5414-4001  |
| 4    | 1    | ALTIMETER PCB (ASSEMBLED)                                            |                         | SPC-5414-5200  |
| 5    | 1    | FREQUENCY SYNTH PCB MkIII (FULLY ASSEMBLED)                          |                         | BCN-1000-5200  |
| 6    | 1    | SPC-5414 (15V) ALTIMETER SEAGLIDER TRANSDUCER WIRING SET (UOW-E4947) | (FROM PLATTER ASSEMBLY) | SPC-5414-4004  |
| 7    | 1    | SEAGLIDER T154 TRANSDUCER ASSEMBLY (UOW-E4947)                       |                         | SPC-5414-7027  |
| 8    | 1    | POWER/COMMS CABLE ASSEMBLY                                           |                         | SPC-5414-4005  |

FOR CLARITY, SOME WIRING ASSEMBLIES ARE NOT SHOWN  
FOR DETAILED INSTRUCTIONS, REFER TO PRODUCTION DOCUMENTATION



ITEM 7  
SEAGLIDER T154  
TRANSDUCER  
ASSEMBLY (UOW-E4947)





## Connections

Refer Figures 2 and 3 below for external component and power connection details. It is recommended external component wiring is kept as short as possible for optimum operation.

See Appendix for transmitter and receiver power connector wiring details.

Figure 2.

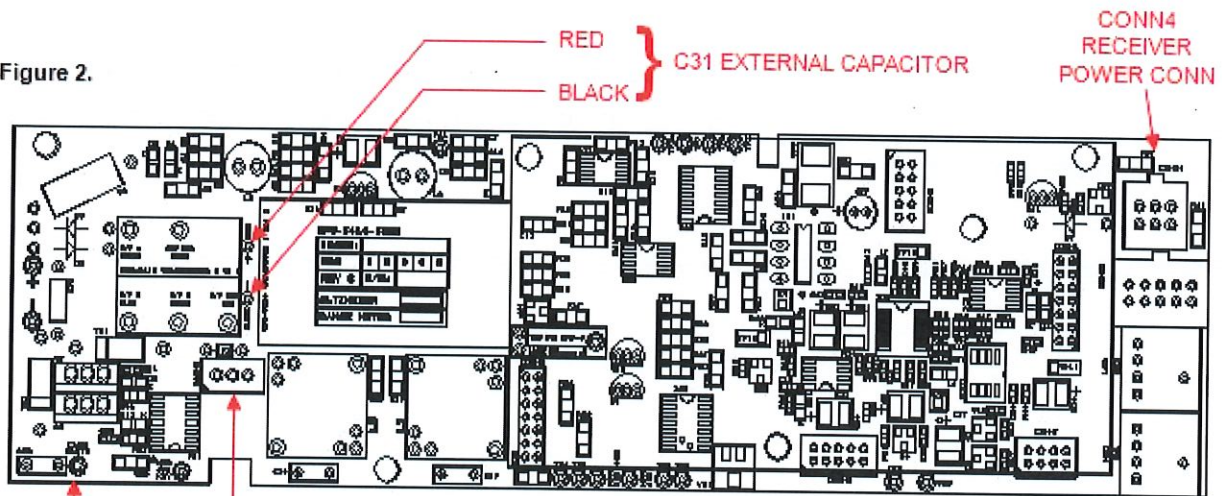
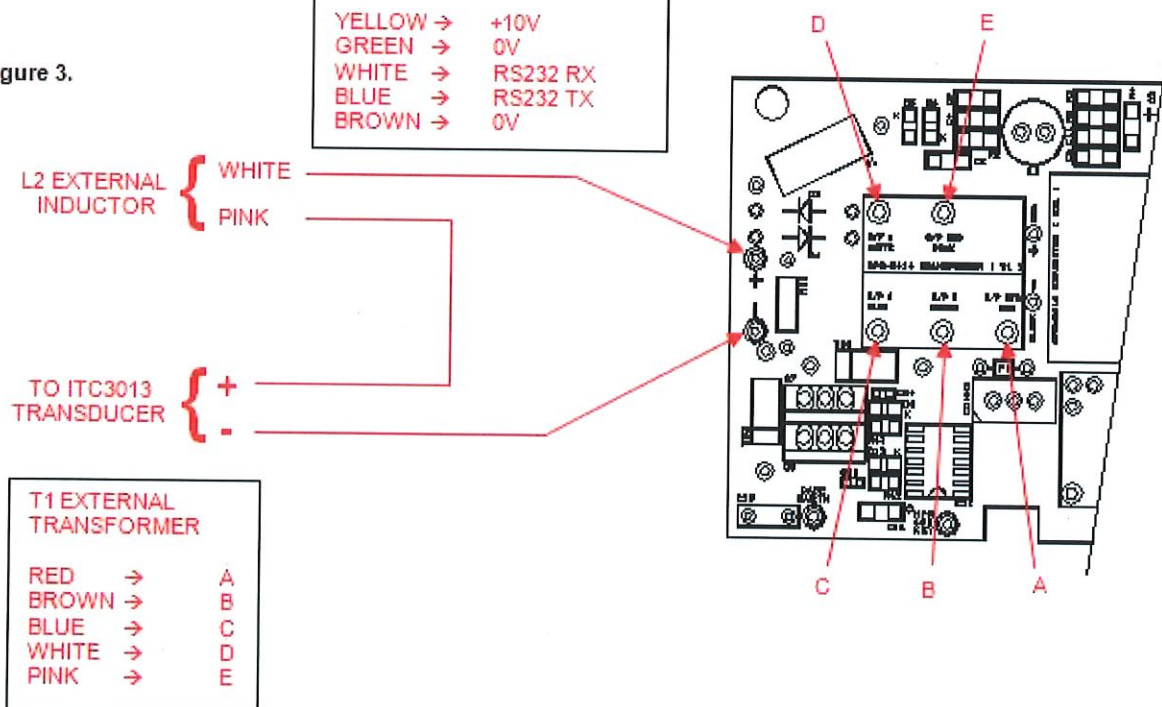


Figure 3.



**Note: DO NOT attempt transponder or altimeter operation if transducer is disconnected.**



**Note: Always test for continuity between transformer ring and clips, and GND point on board.**

**Note: Applied Acoustic Engineering cannot be held responsible for damage or injury caused by incorrect wiring by non AAE personnel.**

#### 4. Transponder Operation



**Note: DO NOT attempt transponder or altimeter operation if transducer is disconnected. Refer section 3.**

## Switching On

After transmitter and receiver power connectors have been inserted, apply specified voltages to the system. Transmitter 24 volts dc and Receiver 10 volts dc. Refer section 3 'System Assembly'.



**Note:** The transponder / altimeter board defaults to transponder mode on power up.

Note: The transponder / altimeter board is fitted with a resettable fuse. If faulty operation of the transponder board is suspected, examination should be undertaken by a competent engineer only. Alternatively, the transponder board set should be de-commissioned and returned to Applied Acoustic Engineering for examination/repair.

## Channel Switches

The two hex rotary switches on the end of the transponder board are used to select a 'channel', i.e. receive and transmit frequencies. These channel selection switches can be rotated freely to the desired position. Note that SW1 is the left hand switch and SW2, the right.



**Note:** During operation, there will be a delay of approximately 60 seconds before a newly selected channel becomes fully processed and operational.

## Special Features

The transponder / altimeter board has the following features: -

## Test Positions

Switch positions 0 / 0 and B / B are used for factory tests only and should not be used by non Applied Acoustic operators.



## Pinger Operation

The board can be set to operate as a free running pinger, as the following table shows.

| Switch position | Repetition rate | Pulse length | Transmit frequency |
|-----------------|-----------------|--------------|--------------------|
| 2 / 8           | 1 Hz            | 10 mS        | 12 kHz             |
| 2 / 9           | 0.5 Hz          | 10 mS        | 12 kHz             |
| 2 / A           | 0.25 Hz         | 10 mS        | 12 kHz             |
| 2 / B           | 0.125 Hz        | 10 mS        | 12 kHz             |
| 4 / 8           | 0.5 Hz          | 10 mS        | 13 kHz             |
| 4 / 9           | 0.25 Hz         | 10 mS        | 13 kHz             |
| 4 / A           | 0.125 Hz        | 10 mS        | 13 kHz             |
| 6 / 8           | 0.5 Hz          | 10 mS        | 15 kHz             |
| 6 / 9           | 0.25 Hz         | 10 mS        | 15 kHz             |
| 6 / A           | 0.125 Hz        | 10 mS        | 15 kHz             |

## Operation with Simrad I507 Systems

The board will operate on the following Simrad I507 system channels as listed in the table below:-

| Switch position | Channel | Interrogate frequency | Reply frequency |
|-----------------|---------|-----------------------|-----------------|
| 0 / 1           | 1       | 9.506 kHz             | 13.158 kHz      |
| 0 / 2           | 2       | 10.309 kHz            | 13.587 kHz      |
| 0 / 3           | 3       | 10.707 kHz            | 14.045 kHz      |
| 0 / 4           | 4       | 11.111 kHz            | 14.451 kHz      |
| 0 / 5           | 5       | 11.547 kHz            | 14.881 kHz      |

The Turn-Around-Time is 30mS.

## Operation with HPR 4 LF

Simrad's low frequency version of their 400 series HPR transmits a double interrogation pulse, using one of the channels listed in the table below:-

| Switch position | Description | Int freq 1 | Int freq 2 | Reply freq |
|-----------------|-------------|------------|------------|------------|
| 1 / 2           | HPR 4 LF    | 10.00 kHz  | 10.50 kHz  | 13.75 kHz  |
| 1 / 3           | HPR 4 LF    | 10.00 kHz  | 11.00 kHz  | 14.25 kHz  |
| 1 / 4           | HPR 4 LF    | 10.00 kHz  | 11.50 kHz  | 14.75 kHz  |
| 1 / 5           | HPR 4 LF    | 10.00 kHz  | 12.00 kHz  | 15.25 kHz  |
| 1 / 6           | HPR 4 LF    | 10.00 kHz  | 12.50 kHz  | 15.75kHz   |
| 2 / 1           | HPR 4 LF    | 10.50 kHz  | 10.00 kHz  | 13.00 kHz  |
| 2 / 3           | HPR 4 LF    | 10.50 kHz  | 11.00 kHz  | 14.00 kHz  |
| 2 / 4           | HPR 4 LF    | 10.50 kHz  | 11.50 kHz  | 14.50 kHz  |
| 2 / 5           | HPR 4 LF    | 10.50 kHz  | 12.00 kHz  | 15.00 kHz  |
| 2 / 6           | HPR 4 LF    | 10.50 kHz  | 12.50 kHz  | 15.50 kHz  |
| 3 / 1           | HPR 4 LF    | 11.00 kHz  | 10.00 kHz  | 13.25 kHz  |
| 3 / 2           | HPR 4 LF    | 11.00 kHz  | 10.50 kHz  | 13.75 kHz  |
| 3 / 4           | HPR 4 LF    | 11.00 kHz  | 11.50 kHz  | 14.75kHz   |
| 3 / 5           | HPR 4 LF    | 11.00 kHz  | 12.00 kHz  | 15.25 kHz  |
| 3 / 6           | HPR 4 LF    | 11.00 kHz  | 12.50 kHz  | 15.75 kHz  |
| 4 / 1           | HPR 4 LF    | 11.50 kHz  | 10.00 kHz  | 13.00 kHz  |
| 4 / 2           | HPR 4 LF    | 11.50 kHz  | 10.50 kHz  | 13.50 kHz  |
| 4 / 3           | HPR 4 LF    | 11.50 kHz  | 11.00 kHz  | 14.00 kHz  |
| 4 / 5           | HPR 4 LF    | 11.50 kHz  | 12.00 kHz  | 15.00 kHz  |
| 4 / 6           | HPR 4 LF    | 11.50 kHz  | 12.50 kHz  | 15.50 kHz  |
| 5 / 1           | HPR 4 LF    | 12.00 kHz  | 10.00 kHz  | 13.25 kHz  |
| 5 / 2           | HPR 4 LF    | 12.00 kHz  | 10.50 kHz  | 13.75 kHz  |
| 5 / 3           | HPR 4 LF    | 12.00 kHz  | 11.00 kHz  | 14.25 kHz  |
| 5 / 4           | HPR 4 LF    | 12.00 kHz  | 11.50 kHz  | 14.75kHz   |
| 5 / 6           | HPR 4 LF    | 12.00 kHz  | 12.50 kHz  | 15.75 kHz  |
| 6 / 1           | HPR 4 LF    | 12.50 kHz  | 10.00 kHz  | 13.00 kHz  |
| 6 / 2           | HPR 4 LF    | 12.50 kHz  | 10.50 kHz  | 13.50 kHz  |
| 6 / 3           | HPR 4 LF    | 12.50 kHz  | 11.00 kHz  | 14.00 kHz  |
| 6 / 4           | HPR 4 LF    | 12.50 kHz  | 11.50 kHz  | 14.50 kHz  |

|       |          |           |           |           |
|-------|----------|-----------|-----------|-----------|
| 6 / 5 | HPR 4 LF | 12.50 kHz | 12.00 kHz | 15.00 kHz |
|-------|----------|-----------|-----------|-----------|

The Turn-Around-Time is 60ms.



**Note:** The SPC-5414-5000 Transponder / Altimeter Board shares electronic circuitry between transponder and altimeter functions. When selecting HPR4 channels with an identical first interrogate frequency, it is recommended channels be chosen with a minimum 1.5KHz spacing between second interrogate frequencies.

### Operation with LF Trackpoint Systems

ORE's Low Frequency Trackpoint systems operates with interrogate frequencies between 9 and 12 kHz and reply frequencies between 12 and 16 kHz, using one of the channels listed in the table below:-

| Switch position | Description | Interrogate frequency | Reply frequency |
|-----------------|-------------|-----------------------|-----------------|
| 0 / 6           | TP2         | 9.00 kHz              | 12.00 kHz       |
| 0 / 7           | TP2         | 9.00 kHz              | 13.00 kHz       |
| 0 / 8           | TP2         | 9.00 kHz              | 14.00 kHz       |
| 0 / 9           | TP2         | 9.00 kHz              | 15.00 kHz       |
| 0 / A           | TP2         | 9.00 kHz              | 16.00 kHz       |
| 1 / 7           | TP2         | 10.00 kHz             | 12.00 kHz       |
| 1 / 8           | TP2         | 10.00 kHz             | 13.00 kHz       |
| 1 / 9           | TP2         | 10.00 kHz             | 14.00 kHz       |
| 1 / A           | TP2         | 10.00 kHz             | 15.00 kHz       |
| 1 / B           | TP2         | 10.00 kHz             | 16.00 kHz       |
| 3 / 7           | TP2         | 11.00 kHz             | 13.00 kHz       |
| 3 / 8           | TP2         | 11.00 kHz             | 14.00 kHz       |
| 3 / 9           | TP2         | 11.00 kHz             | 15.00 kHz       |
| 3 / A           | TP2         | 11.00 kHz             | 16.00 kHz       |
| 5 / 7           | TP2         | 12.00 kHz             | 14.00 kHz       |
| 5 / 8           | TP2         | 12.00 kHz             | 15.00 kHz       |
| 5 / 9           | TP2         | 12.00 kHz             | 16.00 kHz       |



| 7 / 0 | TP2 | 10.50 kHz | 13.00 kHz |
|-------|-----|-----------|-----------|
| 7 / 1 | TP2 | 10.50 kHz | 14.00 kHz |
| 7 / 2 | TP2 | 10.50 kHz | 15.00 kHz |
| 7 / 3 | TP2 | 13.00 kHz | 14.00 kHz |
| 7 / 4 | TP2 | 13.00 kHz | 15.00 kHz |
| 7 / 5 | TP2 | 14.00 kHz | 11.00 kHz |
| 7 / 6 | TP2 | 14.00 kHz | 12.00 kHz |
| 7 / 7 | TP2 | 14.00 kHz | 13.00 kHz |
| 7 / 8 | TP2 | 14.00 kHz | 11.00 kHz |
| 7 / 9 | TP2 | 15.00 kHz | 12.00 kHz |
| 7 / A | TP2 | 15.00 kHz | 13.00 kHz |
| 7 / B | TP2 | 15.00 kHz | 14.00 kHz |
| 8 / 0 | TP2 | 16.00 kHz | 11.00 kHz |
| 8 / 1 | TP2 | 16.00 kHz | 12.00 kHz |
| 8 / 2 | TP2 | 16.00 kHz | 13.00 kHz |
| 8 / 3 | TP2 | 16.00 kHz | 14.00 kHz |
| 8 / 4 | TP2 | 16.00 kHz | 15.00 kHz |

The Turn-Around-Time is 15mS. Select a 10mS transmitted pulse width within the Trackpoint II / II+ system.

The board can be interrogated at 1 PPS (faster than the standard 'Multibeacon').

### Customer Specific Channels

A number of additional channels are available as per customer requirement, listed below:-

| Switch position | Description | Interrogate frequency | Reply frequency |
|-----------------|-------------|-----------------------|-----------------|
| 9 / 0           | Custom      | 12.00 kHz             | 10.50 kHz       |
| 9 / 1           | Custom      | 12.50 kHz             | 10.50 kHz       |
| 9 / 2           | Custom      | 13.00 kHz             | 10.50 kHz       |
| 9 / 3           | Custom      | 14.00 kHz             | 10.50 kHz       |



|       |        |           |           |
|-------|--------|-----------|-----------|
| 9 / 4 | Custom | 14.50 kHz | 10.50 kHz |
| 9 / 5 | Custom | 15.00 kHz | 10.50 kHz |
| 9 / 6 | Custom | 12.00 kHz | 11.50 kHz |
| 9 / 7 | Custom | 12.50 kHz | 11.50 kHz |
| 9 / 8 | Custom | 13.00 kHz | 11.50 kHz |
| 9 / 9 | Custom | 14.00 kHz | 11.50 kHz |
| 9 / A | Custom | 14.50 kHz | 11.50 kHz |
| 9 / B | Custom | 15.00 kHz | 11.50 kHz |
| 9 / C | Custom | 12.00 kHz | 13.50 kHz |
| 9 / D | Custom | 12.50 kHz | 13.50 kHz |
| 9 / E | Custom | 13.00 kHz | 13.50 kHz |
| 9 / F | Custom | 14.00 kHz | 13.50 kHz |
| A / 0 | Custom | 14.50 kHz | 13.50 kHz |
| A / 1 | Custom | 15.00 kHz | 13.50 kHz |

The Turn-Around-Time is 20mS and transmitted pulse width is 10mS.

## 5. Altimeter Operation



**Note: DO NOT attempt transponder or altimeter operation if transducer is disconnected. Refer section 3.**

### Switching On

After transmitter and receiver power connectors have been inserted, apply specified voltages to the system. Transmitter 24 volts dc and Receiver 10 volts dc. Refer section 3 'System Assembly'.



**Note:** The transponder / altimeter board defaults to transponder mode on power up.

Note: The transponder / altimeter board is fitted with a resetable fuse. If faulty operation of the transponder board is suspected, examination should be undertaken by a competent engineer only. Alternatively, the transponder board set should be de-commissioned and returned to Applied Acoustic Engineering for examination/repair.

## Altimeter Description

The altimeter detector is implemented using a variable gain amplifier and an envelope detector, deriving its input from the transponder receiver. On transmitting a frequency pulse, the gain of the variable gain amplifier is controlled to compensate for the attenuation in water with distance of the transmitted pulse. If the transponder receiver is subsequently triggered, the envelope detector output is analysed in software using a user selectable sensitivity setting. Based on transmitted pulse width, this validates the received signal as a legitimate reflection, or ignores the received signal as noise.

The maximum range is limited to 500 meters (1000 meters total distance), equivalent to 667 milliseconds.

## Serial Protocol

A simple RS232 command protocol is used to control board operation as an altimeter.

Serial communication settings:      9600 baud rate  
                                                 8 data bits  
                                                 1 stop bit  
                                                 No parity  
                                                 No flow control

A total of 16 serial commands are implemented.

Inverted logic levels are used rather than true RS232 levels, i.e. mark is a '0' and space is a '1'. Inputs and outputs are 5 volt TTL / CMOS compatible.

**The onboard microcontroller requires an arbitrary character to 'wake up' from its quiescent transponder state. Note this is applicable on first power-up and after a return to transponder mode.**

If no serial activity is detected for 2 minutes, the board will revert back to its quiescent transponder mode. Note an arbitrary character received within the 2 minute window will 'reset' this window, i.e. a command is not necessary.

With all serial commands, a successful change will reply with OK <CRLF>, or if trying to assign a new value out of range with INVALID <CRLF>. Note replies to serial commands can be suppressed if required.



## Serial Commands

- **To enable altimeter function**

\$An <CRLF>

Where

A is the altimeter ON command,

n is the repetition rate in seconds from 0 – 9.

If n is 0, single ping only.

Note that after a single ping, the microcontroller is still active and power consumption will be approximately 7 milliamps.

Note during altimeter function, power consumption can reach a maximum of approximately 35 milliamps during the altimeter time-out window, depending on received signal input.

- **To adjust the sensitivity of the altimeter detector**

\$Sn <CRLF>

Where

S is the sensitivity function,

n is the sensitivity figure between 0 – 5.

The default setting is 3 or the last setting used (stored in flash memory). The lowest sensitivity setting is 0 and the highest is 5.

Note refer altimeter sensitivity adjustment section for more details.



- **To set the altimeter lock-out (minimum range)**

\$Ln <CRLF>

Where

L is the lock-out at the end of the transmit pulse width,  
n is the time in milliseconds from 0 – 9.

The default setting is 2 milliseconds or the last setting used (stored in flash memory).

- **To set the altimeter time-out (maximum range)**

\$Mxxx <CRLF>

Where

M is the time-out of the altimetry function,  
xxx is the time in milliseconds from 001 – 667 (for an altimeter range of 500 meters).

The default setting is 667 milliseconds or the last setting used (stored in flash memory).

- **To set the altimeter transmit frequency**

\$Fyy <CRLF>

Where

F is the altimeter transmit frequency,  
yy is the frequency from 10 – 25 KHz in 1 KHz steps.

The default setting is 12KHz or the last setting used (stored in flash memory).

- **To set the altimeter transmit pulse width**

\$Pz <CRLF>

Where

P is the altimeter transmit pulse width,

z is the pulse width in milliseconds from 1 – 9.

The default setting is 1 millisecond or the last setting used (stored in flash memory).

- **To enable / suppress serial command replies**

\$Cn <CRLF>

Where

C is the serial command suppress control,

n is 0, replies are suppressed,

n is 1, replies are enabled.

The default setting is enabled or the last setting used (stored in flash memory).

- **To switch to transponder mode**

\$T <CRLF>

Where

T is the transponder ON command.

The board will operate as a transponder set by the two channel select rotary switches on the board. Power consumption will drop to approximately 2 milliamps.

- **To switch to transponder mode with user-selected transponder frequencies.**

\$Tff.f,rr.r <CRLF>

Where

T is the transponder ON command and,

ff.f is the transponder interrogate frequency 08.0 to 16.0kHz in 0.1kHz steps,

rr.r is the transponder reply frequency 08.0 to 16.0kHz in 0.1kHz steps.

The board will operate as a transponder using user-selected interrogate and reply frequencies. Power consumption will drop to approximately 2 milliamps.

Note turn-around-time and transmit pulse width are determined by rotary switch settings.

Note transponder mode defaults to rotary switch settings interrogate and reply frequencies on power up.

- **To query transponder channel set by the channel select rotary switches**

\$R <CRLF>

Where

R is the query transponder channel set by rotary switches.

Transponder channel is transmitted serially in format,

xx <CRLF> where xx is hexadecimal 00 to FF.

Transponder channel is always determined by rotary switch settings regardless of current transponder interrogate and reply frequencies.



Note turn-around-time and transmit pulse width are determined by rotary switch settings.

- **To query current transponder interrogate and reply frequencies**

\$V <CRLF>

Where

V is the query current transponder interrogate and reply frequencies command.

Interrogate and reply frequencies are transmitted serially in the format, ff.f,rr.r <CRLF> where,

ff.f is the transponder interrogate frequency 08.0 to 16.0kHz to nearest 0.1kHz,  
rr.r is the transponder reply frequency 08.0 to 16.0kHz to nearest 0.1kHz,

Current transponder interrogate and reply frequencies are determined by the command previously used to switch to transponder mode i.e.,

\$T command, rotary switch settings interrogate and reply frequencies.

\$Tff.f,rr.r command, user-selected interrogate and reply frequencies.

Note turn-around-time and transmit pulse width are determined by rotary switch settings.

- **To control transponder interrogate function**

\$Bn <CRLF>

Where:

B is the control transponder interrogate function command and,

n is 0 for interrogate function enabled,

n is 1 for interrogate function disabled.

If the interrogate function is disabled, after switching to transponder mode the board will remain in its quiescent power state consuming approximately 2 milliamps until a serial character is received.

### **To query interrogate function status**

\$B <CRLF>

Interrogate function status is transmitted serially in format,  
n <CRLF>

Note interrogate function is always enabled on power up.

### **To change transponder receive inhibit time**

\$Ixx <CRLF>

Where

I is the inhibit time,

xx is the time in 100 millisecond steps from 01 – 99.

The default setting is 800 milliseconds or the last setting used (store in flash memory).

- **To query number of transponder interrogations**

\$N <CRLF>

Where

N is the query number of transponder interrogations command.

Count is transmitted serially in the format,

Cxxxxx <CRLF> where xxxxx is the count 00000 to 65535.

Count is reset to zero after viewing.

Note after 'wake up', if viewing of the current transponder interrogation count is not required, any transponder interrogation count additions during 'wake up' will be retained correctly in flash memory only if the user returns to the quiescent transponder mode by issuing \$T command. If the board is

power cycled in 'wake up', any transponder interrogation count additions during 'wake up' period will be lost.

- **To query firmware revision**

\$W <CRLF>

Where

W is the query firmware revision command.

Firmware is transmitted serially as a character string e.g.,  
3.08 E4947 <CRLF>.



## **User ID strings**

ID strings can be used to identify board set hardware as required by the user.

A total of 4 alphanumeric strings of 1 to 8 characters long can be stored.

- **To read ID string**

\$HRn <CRLF>

Where

HR is read ID string command,  
n is string 1 to 4.

ID string is transmitted serially in the format,  
aaaaaaaa <CRLF> where aaaaaaaaa is alphanumeric string 1 to 8  
characters long.

Default ID string is 00000000.

- **To write ID string**

\$HWn,aaaaaaaa <CRLF>

Where

HW is write ID string command,  
n is string 1 to 4 (note proceeding comma),  
aaaaaaaa is alphanumeric string 1 to 8 characters long.

Strings are stored in flash memory.

- **To adjust the level of spurious transponder trigger rejection**

\$Zn <CRLF>

Where

Z is the spurious transponder trigger rejection function

n is the rejection level 0 – 6

A transponder interrogate trigger event is monitored over a 10mS period after triggering. The trigger is accepted if the total time the trigger is legal over this 10mS period, i.e trigger length, is equal to or greater than the selected rejection level.

**Rejection Level n Trigger Length (Over 10mS Period)**

|   |                   |
|---|-------------------|
| 6 | >= 3.00mS         |
| 5 | >= 2.50mS         |
| 4 | >= 2.00mS         |
| 3 | >= 1.50mS         |
| 2 | >= 1.00mS         |
| 1 | >= 0.50mS         |
| 0 | Triggers directly |

The default setting is 0 or the last setting used (stored in flash memory). The lowest rejection level is 0 and the highest is 6.

**Altimeter data out**

Altimeter detect times are transmitted serially in the format,

Rxxx.xxxms <CRLF> where xxx.xxx is the 2-way travel time in milliseconds to three decimal places.

If no valid reflection is detected during the time-out window,

R----- <CRLF> is transmitted.

Valid detect times are accurate to within  $\pm 100$  microseconds resolution of a received reflection.

### Altimeter Sensitivity Adjustment

Over the altimeter operational frequency range of 10 – 25 KHz, transmit source-pressure levels and receive sensitivity will vary greatly. Also, due to utilising an arbitrary transducer with this board set and external components, transmit source-pressure levels and transducer receive sensitivities will again vary, between individual transducers. To help compensate for this significantly wide variety of operating conditions, a method of adjusting the altimeter detector circuitry output presented for software analysis, has been implemented.

The altimeter detector, as previously described, comprises a variable gain amplifier and an envelope detector. The altimeter detector input is derived from the transponder receiver. On transmitting a frequency pulse, the gain of the variable gain amplifier is controlled to compensate for the attenuation in water with distance of the transmitted pulse. If the transponder receiver is subsequently triggered, the envelope detector output is analysed in software using a user selectable sensitivity setting. Based on the transmitted pulse width, this validates the received signal as a legitimate reflection, or ignores the received signal as noise.

The transponder receiver comprises a tuned input stage followed by two operational amplifier stages, to split the receiver gain. The altimeter detector derives its input from the output of the first operational amplifier stage, having a fixed gain of 23.5dB, which can be monitored at test point TP2. This also gives the user an amplified input signal to digitise if required at this test point.

The gain of the variable gain amplifier is controlled using an approximation to a logarithmic curve characteristic, to compensate for attenuation in water of the transmitted pulse, due to spreading and absorption. This gain control curve is fixed, modelled on the ideal attenuation curve of a 12KHz transmit pulse, varying by a maximum deviation of 5dB from the ideal. The maximum gain of the altimeter detector is 48dB, giving effective compensation for attenuation up to a range of 100 meters (200 meters total distance). If the speed of sound in seawater is approximately 1500 meters / second, gain will reach its maximum of 48dB for a range of 100 meters in 133 milliseconds. Though the gain is effectively a maximum at 100 meters range, the detector software sensitivity setting can be reduced by



the user to enable ranging at longer distances up to the maximum of 500 meters (total distance 1000 meters and 667 milliseconds maximum time-out window).

The output of the variable gain amplifier is input to the envelope detector, to recover the pulse shape and width of an expected reflection. The gain of the envelope detector can be manually adjusted by on-board potentiometer VR1, adjusting the output presented to the microcontroller for software sensitivity analysis. The output of the envelope detector is a maximum 5 volts, so too much gain will clip the received signal.

As stated previously, software analysis of the envelope detector output validates a legitimate reflection from the user selected software sensitivity level and the transmitted pulse width. Basically, the sensitivity setting represents a limit the output of the envelope detector must exceed for the duration of the transmitted pulse time. Software sensitivity equates to D.C. (direct current) levels as follows:

| Sensitivity | D.C. level                            |
|-------------|---------------------------------------|
| 5           | 4.00 volts                            |
| 4           | 2.00 volts                            |
| 3           | 1.00 volts                            |
| 2           | 0.50 volts                            |
| 1           | 0.25 volts                            |
| 0           | Altimeter detector circuitry not used |

Current consumption of the board set when using the altimeter detector circuitry can **peak at a maximum of 35 milliamps over the duration of the time-out window**. A setting of '0' will not use the altimeter detector circuitry, and hence software analysis, triggering purely on a signal at the transponder receiver output. This saves significantly on power consumption, **down to a peak of 12 milliamps**, but obviously could trigger on spurious received noise.

Unfortunately, adjusting the output of the envelope detector correctly is dependent on multiple factors. With an arbitrary transducer and depending on the frequency used, transmit source-pressure level and receive sensitivity will vary enormously. Also the gain control curve approximation to an ideal logarithmic curve will limit absolute accuracy.

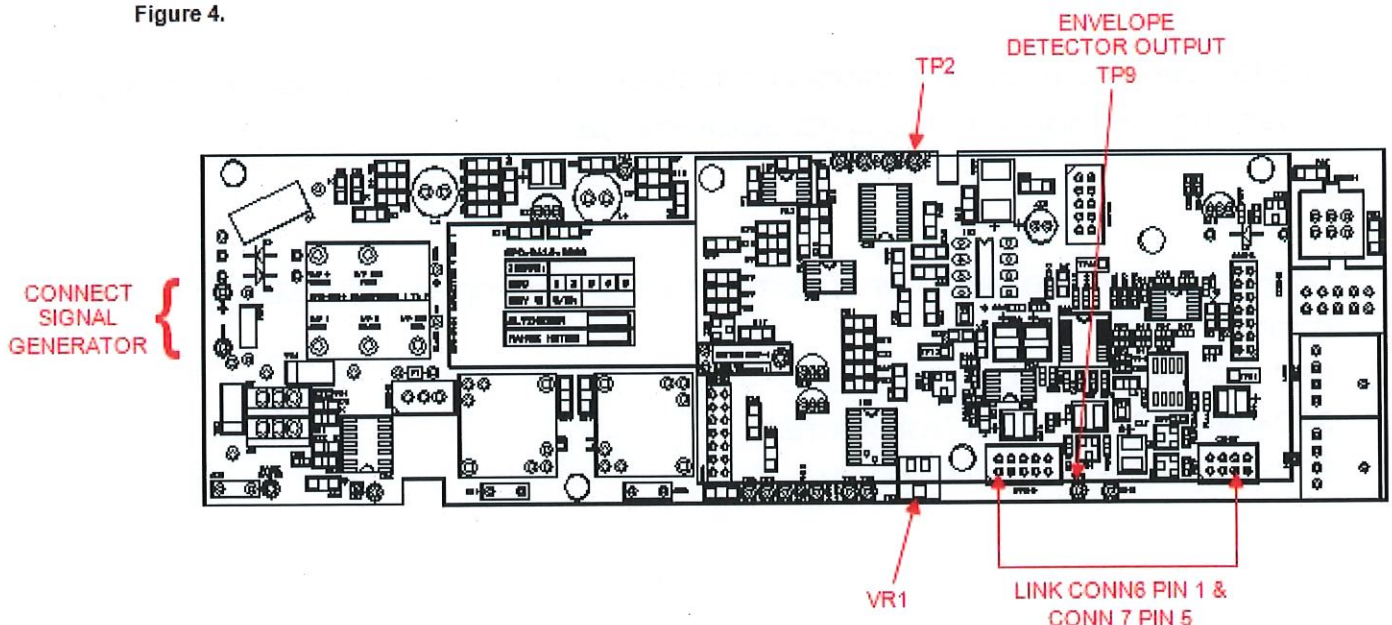
As an example in setting up, using an operational frequency of 12 KHz:

- Assumed maximum source-pressure level @ 12 KHz = 191dB
- Receive sensitivity of an arbitrary ITC3013 transducer @ 12 KHz = -181dB maximum  
(Derived from a sample spread of ITC3013 datasheets).
- Transmit pulse width = 9 milliseconds
- Altimeter gain a maximum of 48dB @ 100 meters range = 200 meters total distance
- Assume 100% reflected power

From these parameters, and accounting for spreading and absorption, it is expected an approximate 44 millivolts peak-to-peak received sinusoidal signal would be developed across the transducer for a range of 100 meters.

An oscilloscope and a signal generator with tone-burst capability are required. Remove the processor daughterboard and connect points as indicated on Figure 4 below:

Figure 4.





Note the link between Conn 6 Pin 1 and Conn 7 Pin 5 enables the altimetry detector circuitry during this example.

**Connect power to the board using receiver power connector at CONN4 only. Do not connect power until all connections as Figure 4 have been made.**

Set the signal generator to output a 9 millisecond tone burst with a 1 second repetition rate. Monitor test point TP2 on the oscilloscope. This is the output of the first operational amplifier stage of the transponder receiver, with a gain of 23.5dB. This gain in dB is equivalent to an amplification factor of 15. Therefore adjust the output amplitude of the signal generator until the tone burst monitored on TP2 is approximately 660 millivolts peak-to-peak (15 multiplied by 44 millivolts peak-to-peak).

Monitor the envelope detector output at TP9 on the oscilloscope. The signal envelope will follow the 9 millisecond toneburst. As have assumed 100% reflected power, adjust VR1 until the complete top edge of the envelope is above the maximum software sensitivity equivalent limit of 4.00 volts.

Set up is now complete.

## 6. Spares

The following spares are available from stock. A part number is not necessary to order. The description below is acceptable to us.

### Components

SPC-5414-5000 Transponder / Altimeter board (complete with MkIII 'processor board).

SPC-5414-3000 External Inductor

SPC-5414-3001 External Transformer

SPC-5414-4001 External Capacitor

### Connector Parts

Transponder Switch Control Cable assembly

Please quote SPC-5414-5000 when ordering spares.



## 7. Fault Finding

### **It doesn't work, what can we do ?**

There are two primary questions here which will help to identify the nature of the problem. Most instances of failure are due to mechanical damage or 'finger trouble' but the possibility of component failure is always there.

### **Did it ever work?**

If the answer is yes, either the unit is set-up incorrectly or it is faulty. Look for mechanical damage.

If used as a transponder, has the correct channel been selected on the board?

If used as an altimeter, are the correct serial comms settings selected?

There is an 8-way DIL switch on the 'processor board, SW1. Has someone knocked them? All switches should be in the OFF position.

If these tests do not help or resolve the problem, contact Applied Acoustic Engineering immediately for technical assistance.

### **Is performance poor?**

- If the answer is yes, then please consider the following:-
- If you have a similar model, is the performance the same? This will identify if there is just one faulty unit.
- Are you out of range? Again check with a similar model. Maybe a higher powered unit is required.
- Is multipath present? Try slowing down the interrogation rate.

- Are you within the beam pattern of the transducer? If the signal from the beacon is not illuminating the vessel, then it cannot be tracked. Cable catenary has been known to cause problems occasionally with tow fish tracking (the signal may transmit at the wrong angle and will miss the ship).
- If you are tow fish tracking and there is a large water temperature change between the surface of the water and the beacon position, you may be experiencing severe ray bending.
- If results are different to another model? You may be comparing different models and different specifications.
- Check on a different channel (frequency) to see if performance changes.

If you can't find a solution, please contact us. Call, fax, write or e-mail and we will try to help wherever we can. Contact details are at the back of this manual. Address your communication to Technical Support.

## 8. Specifications

### General Specifications

|                                  |                                                                                                   |
|----------------------------------|---------------------------------------------------------------------------------------------------|
| Model No.                        | SPC-5414-5000                                                                                     |
| Transmitter Power                | 191dB $\pm$ 3dB with AAE T154 transducer.                                                         |
| Receiver step size               | 32 Hz / 32 Hz                                                                                     |
| Transmitter step size            | 32 Hz / 250 Hz                                                                                    |
| Power supply                     | Receiver 10 volts dc<br>Transmitter 15 volts dc                                                   |
| RS232 electrical characteristics | 5V TTL / CMOS compatible<br>Inverted logic, mark is '0', space is '1'                             |
| Serial commands                  | 16 in total.                                                                                      |
| Serial comms settings            | 9600 baud rate<br>8 data bits<br>1 stop bit<br>No parity<br>No flow control                       |
| Board dimensions                 | 192mm x 55mm x 20mm (L x W x H).                                                                  |
| External inductor                | Part No. SPC-5414-3000<br>9mH.<br>Dimensions 45mm diameter x 28mm high.                           |
| External transformer             | Part No. SPC-5414-3001<br>Primary 9+9t / Secondary 120t.<br>Dimensions 35mm diameter x 24mm high. |
| External capacitor               | Part No. SPC-5414-4001<br>22000uF 35v<br>Dimensions 30mm diameter x 50mm long.                    |
| Operating temperature            | -10 to 30 C                                                                                       |
| Storage temperature              | -10 to 45 C                                                                                       |





## Transponder Specifications

|                        |                                                   |    |
|------------------------|---------------------------------------------------|----|
| Frequency Band         | 10 – 16 KHz.                                      |    |
| Expected useable range | 3000m                                             |    |
| Turn around delay      | Dependent on system selected.                     |    |
| Channels / frequencies | HPR 1507                                          | 5  |
|                        | HPR 4 LF                                          | 30 |
|                        | ORE Trackpoint II                                 | 34 |
|                        | Pinger                                            | 10 |
|                        | Custom                                            | 18 |
| Spare channels         | Yes                                               |    |
| Channel selection      | Two hex rotary switches.                          |    |
| Inhibit time           | User selected by serial command<br>100ms – 9900ms |    |

## Altimeter Specifications

|                         |                                                                      |
|-------------------------|----------------------------------------------------------------------|
| Transmit frequency band | 10 – 25 KHz                                                          |
| Transmitted pulse width | 1 – 9 ms                                                             |
| Receive timeout         | 0 – 667ms                                                            |
| Receive lock-out        | 0 – 9ms                                                              |
| Sensitivity             | 6 software sensitivity settings.<br>Hardware sensitivity adjustable. |
| Detector maximum gain   | 48dB                                                                 |

Gain control

Fixed approximation to logarithmic curve to compensate for attenuation of 12 KHz transmit pulse.

5dB.

Maximum deviation from ideal logarithmic curve

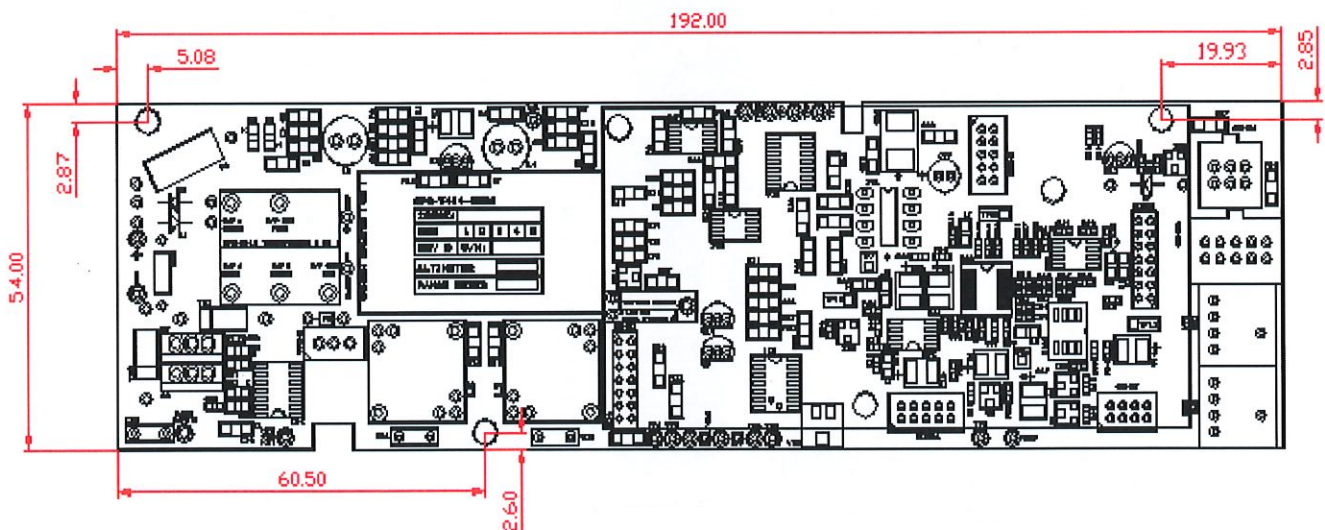
Maximum gain attained after 133ms.

Detection accuracy

$\pm 100\mu s$

## 9. Appendix

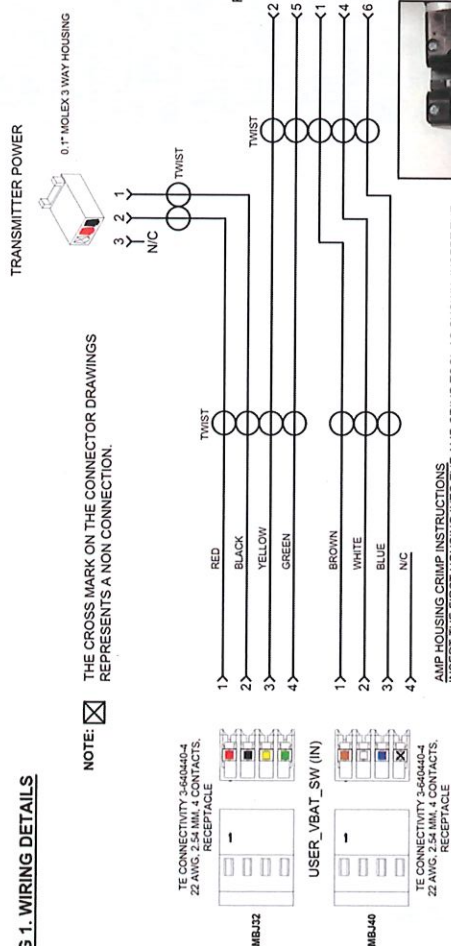
### Board Dimensions



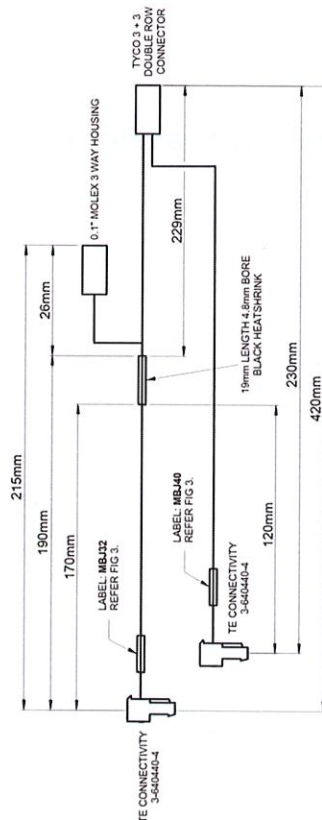
Note: Max height is 20mm.

AAE Ref: SPC-5414-4005-1

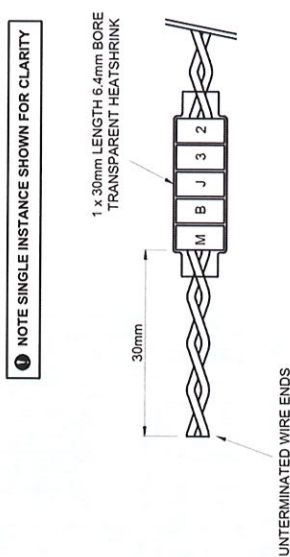
**NOTE:** 



**FIG 2. WIRING LENGTHS**



**FIG 3. LABELLING DETAILS**







Applied Acoustic Engineering Limited is a leading company in the design and manufacture of a wide range of subsea navigation and positioning products, and marine seismic survey equipment.

The extensive product range includes the innovative USBL tracking system, Easytrak, a variety of positioning and release beacons and seismic sub-bottom profiling equipment for offshore geotechnical and seabed analysis.

All products use acoustics, underwater sound waves, in location, positioning, navigation and data acquisition applications.